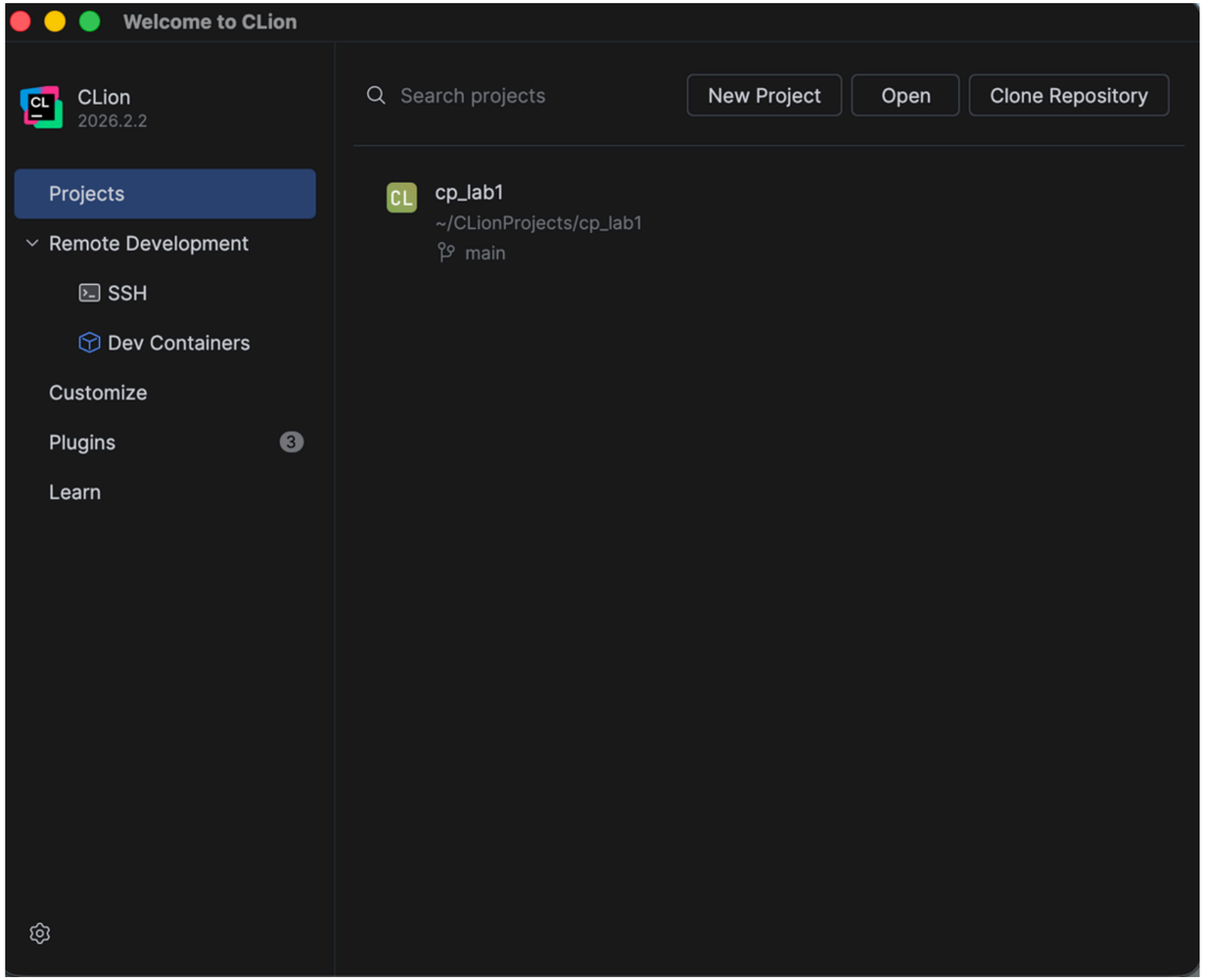
COMPUTER PROGRAMMING WEEK 1 LAB 2

New Uzbekistan University

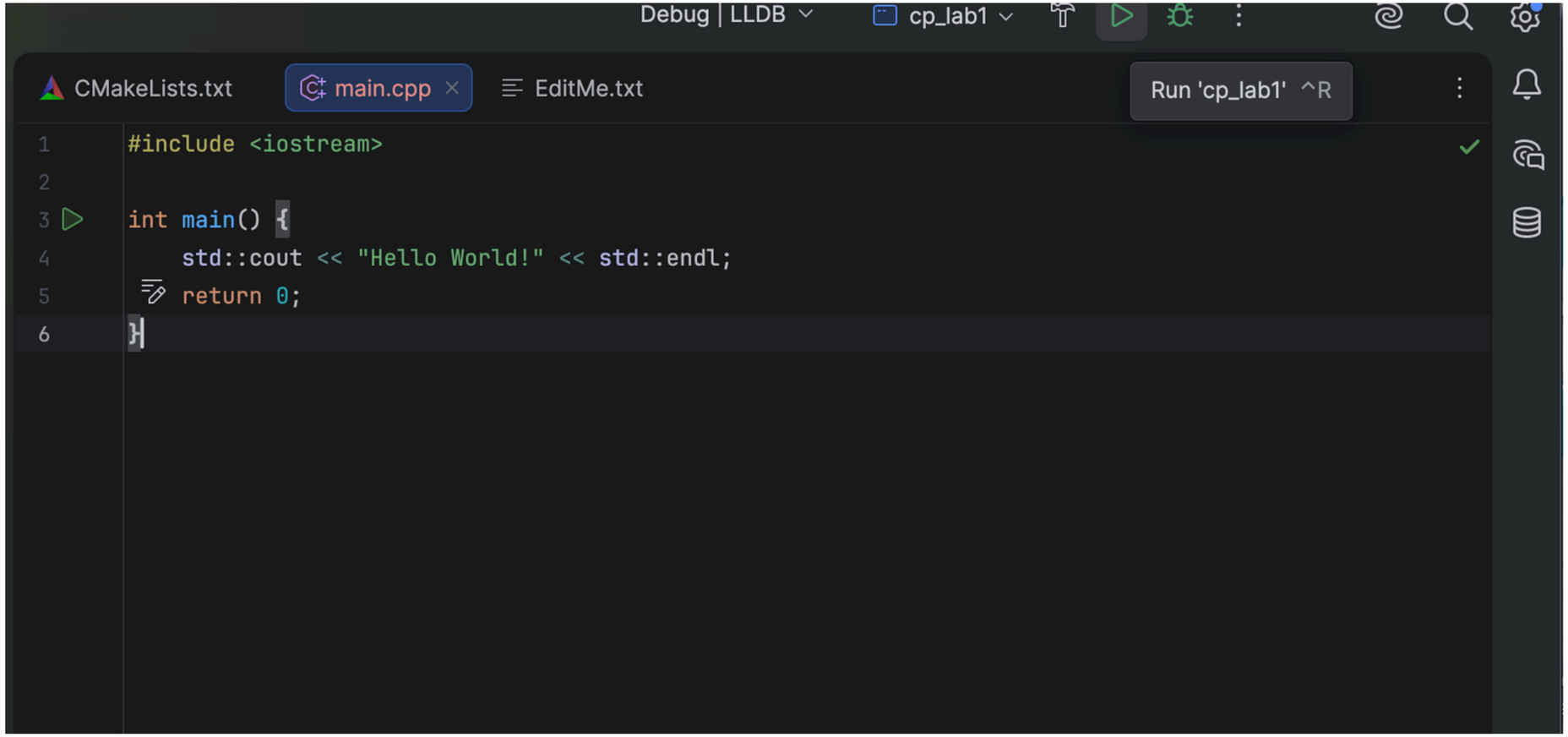
Sarvinoz Anvarova

Open CLion



Open our cp_lab1 project which we created in the last lab

Write the code



Go to main.cpp file and write “Hello world” code. Run it using the toolbar above

View the output



You should see this output after running the code succesfully

Check git status

```
(base) sarvinoz@Mac cp_lab1 % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .idea/
        main.cpp

nothing added to commit but untracked files present (use "git add" to track)
```

It shows that we didn’t add main.cpp to the git yet

Adding main.cpp to git

```
(base) sarvinoz@Mac cp_lab1 % git add main.cpp
```

```
(base) sarvinoz@Mac cp_lab1 % git diff --staged
diff --git a/main.cpp b/main.cpp
new file mode 100644
index 0000000..470a39d
--- /dev/null
+++ b/main.cpp
@@ -0,0 +1,6 @@
+#include <iostream>
+
+int main() {
+    std::cout << "Hello World!" << std::endl;
```

Use **git add main.cpp** command to add main.cpp file to git. You can see the difference made using **git diff --staged** command or use **git status** as before

Note: to escape git diff --staged output use “q” key

Commit main.cpp

```
(base) sarvinoz@Mac cp_lab1 % git commit -m "Add Hello World program"
[main a23d73d] Add Hello World program
 1 file changed, 6 insertions(+)
 create mode 100644 main.cpp
```

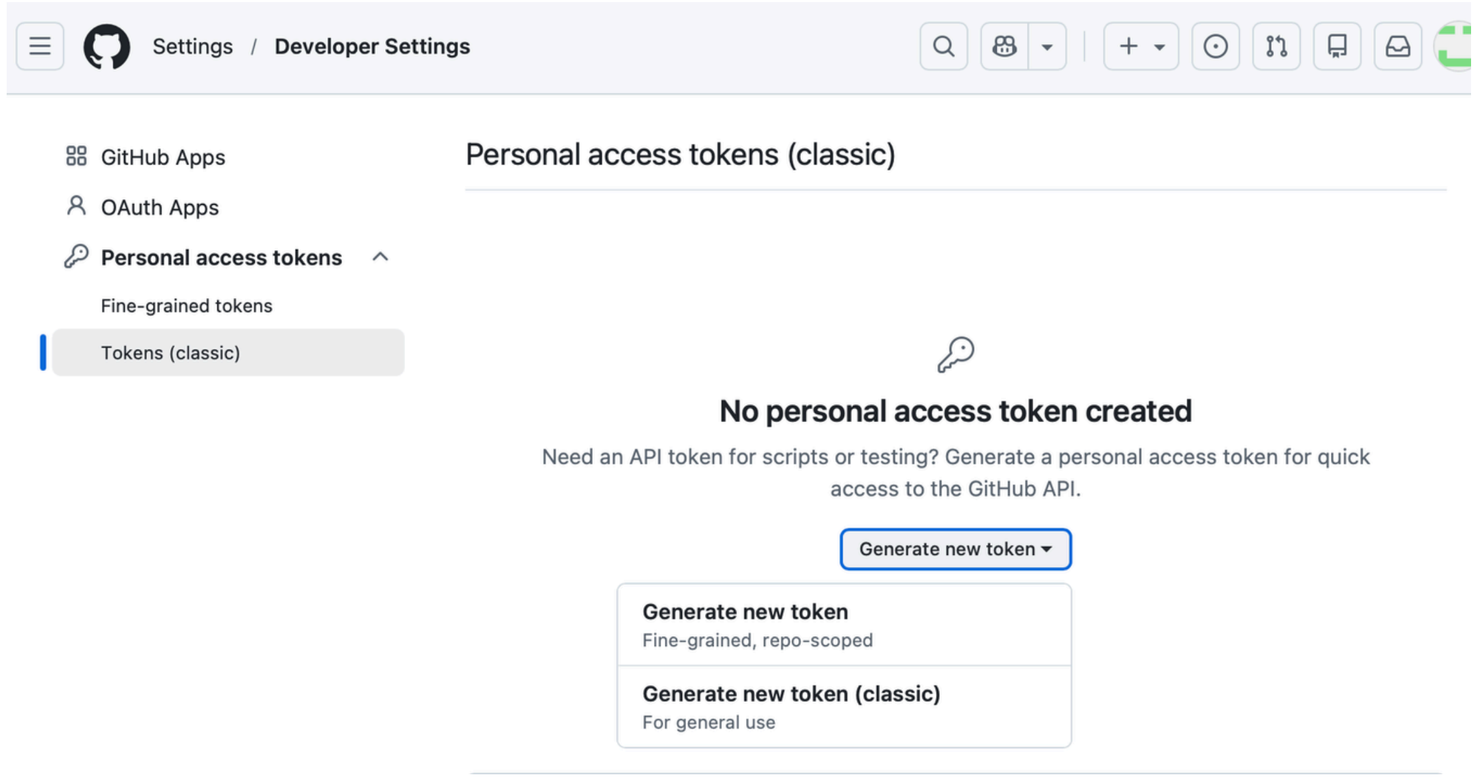
Note: CMakeLists.txt is already in the repository, so do not add it again unless CLion modified it. If you didn’t add CMakeLists.txt file in the previous lab please add it.

Push main.cpp 1st method

```
(base) sarvinoz@Mac cp_lab1 % git push
Username for 'https://github.com': srvinn1
Password for 'https://srvinn1@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 411 bytes | 411.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/srvinn1/cp_lab1.git
   5a13ffc..a23d73d  main -> main
```

You can push using different methods: 1st way is to push in terminal using **git push** command. But for that you need to have a token in github

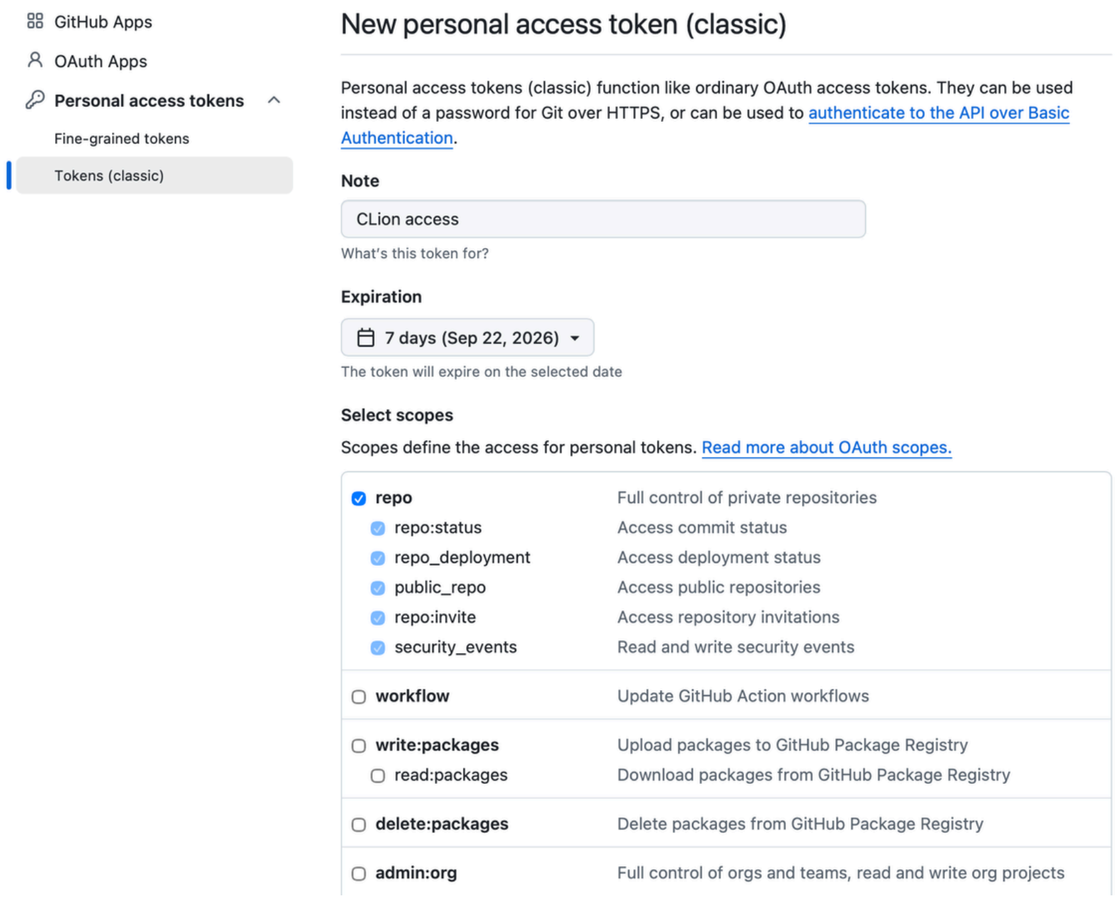
Obtaining Github token (optional)



To obtain token go to Settings → Developer Settings → Personal acess tokens → Tokens (classic)

Select Generate new token (classic)

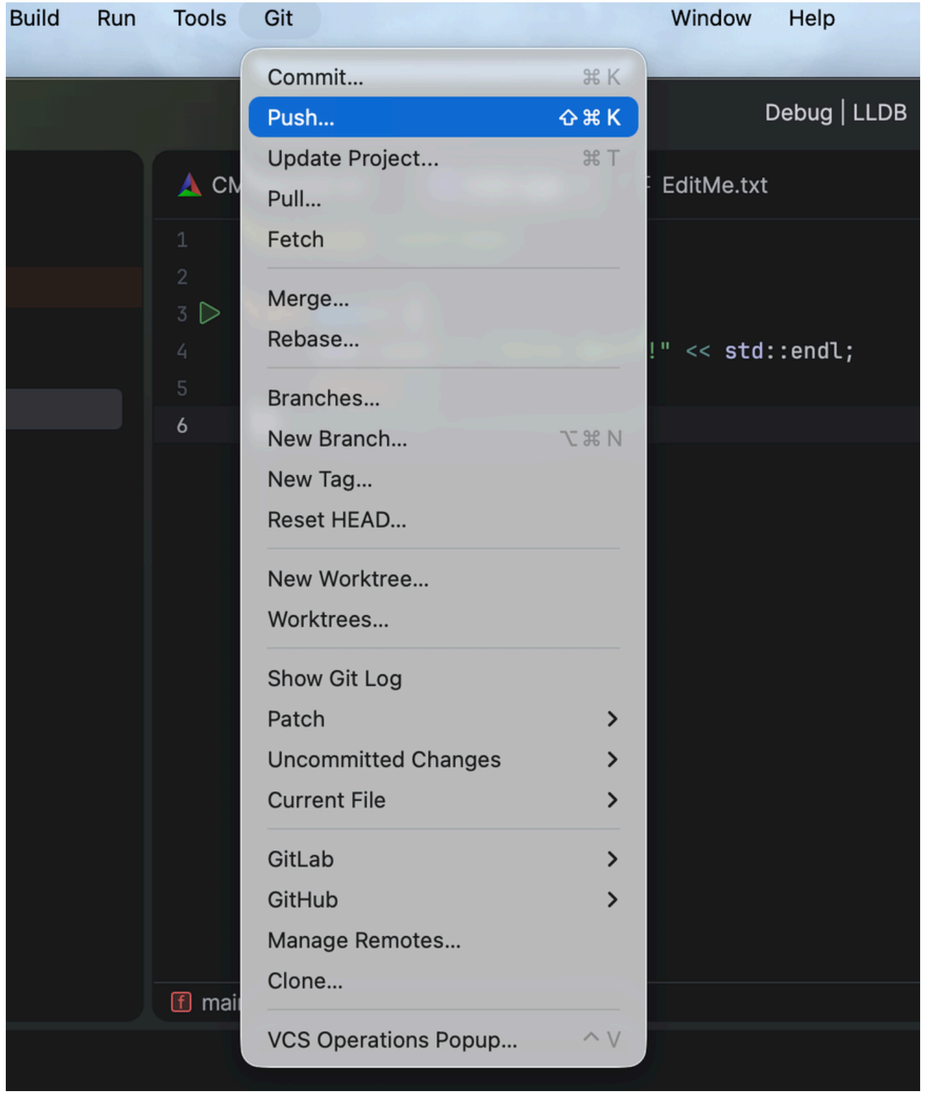
Obtaining Github token (optional)



Select everything as shown here and generate the token. Copy it since it will be shown only once.

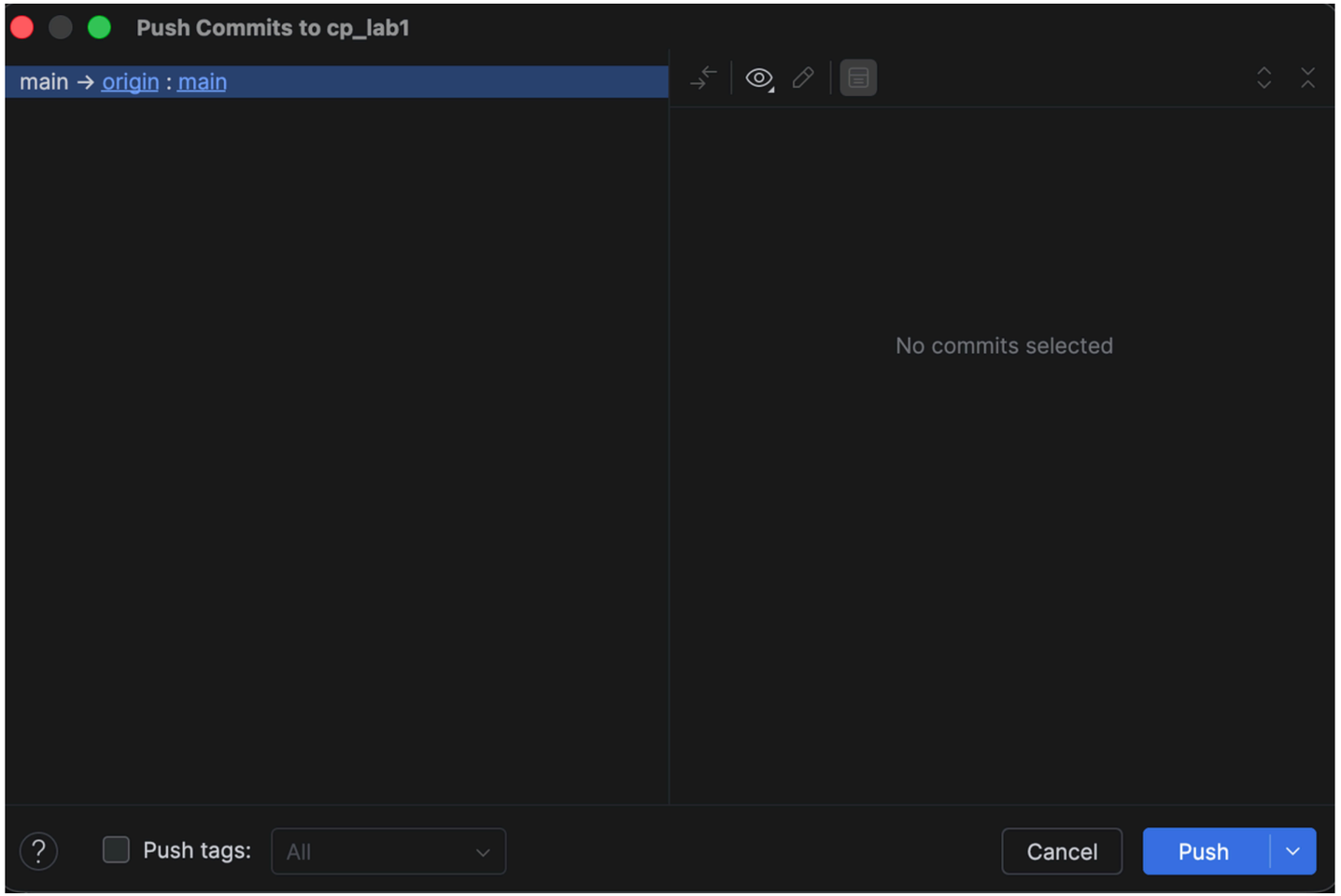
DO NOT share your tokens! It is personal and meant not to be shared

Push main.cpp 2nd method



Go to Git → Push

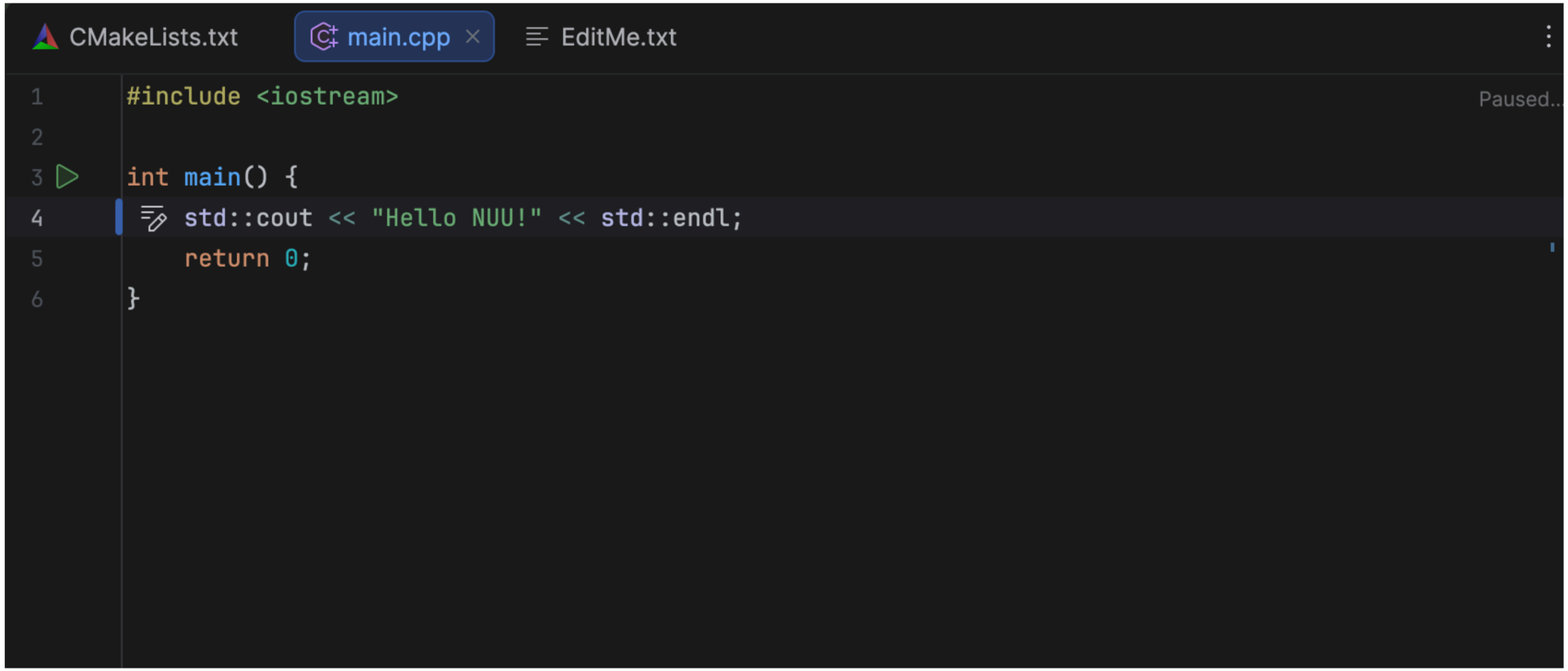
Push main.cpp 2nd method



Press “Push”

For this method you do not need to have a token

Change main.cpp



Change your code from “Hello world” to “Hello NUU”

Run and Save your files

Inspect the change using git

```
(base) sarvinoz@Mac cp_lab1 % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   main.cpp

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .idea/
```

Use git status to verify the changes

You can also run **git diff main.cpp** to see which changes were applied to main.cpp file

```
(base) sarvinoz@Mac cp_lab1 % git diff main.cpp
diff --git a/main.cpp b/main.cpp
index 470a39d..0ce3274 100644
--- a/main.cpp
+++ b/main.cpp
@@ -1,6 +1,6 @@
 #include <iostream>

 int main() {
-    std::cout << "Hello World!" << std::endl;
+    std::cout << "Hello NUU!" << std::endl;
     return 0;
:
```

Adding changed main.cpp to git

```
(base) sarvinoz@Mac cp_lab1 % git add main.cpp
```

Use git add main.cpp command again to add main.cpp to git

Commit and push the changes

```
(base) sarvinoz@Mac cp_lab1 % git commit -m "Change greeting to Hello NUU"
[main bb710df] Change greeting to Hello NUU
 1 file changed, 1 insertion(+), 1 deletion(-)
```

main.cpp was committed and pushed successfully

```
(base) sarvinoz@Mac cp_lab1 % git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 304 bytes | 304.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/srvinn1/cp_lab1.git
   a23d73d..bb710df  main -> main
```

View all our commits

```
(base) sarvinoz@Mac cp_lab1 % git log --oneline
bb710df (HEAD -> main, origin/main) Change greeting to Hello NUU
a23d73d Add Hello World program
5a13ffc First commit
```

You can see all our commits!

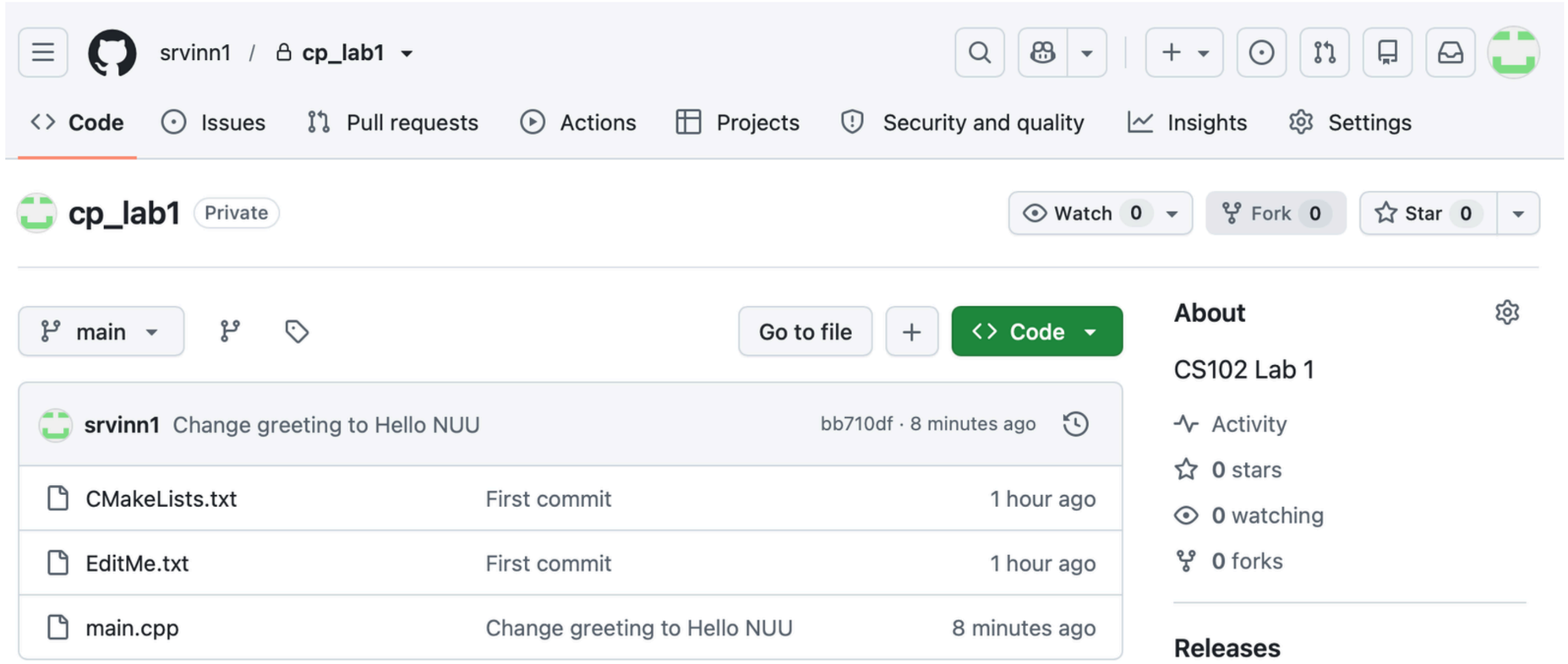
View all our commits

```
(base) sarvinoz@Mac cp_lab1 % git log --oneline
bb710df (HEAD -> main, origin/main) Change greeting to Hello NUU
a23d73d Add Hello World program
5a13ffc First commit
```

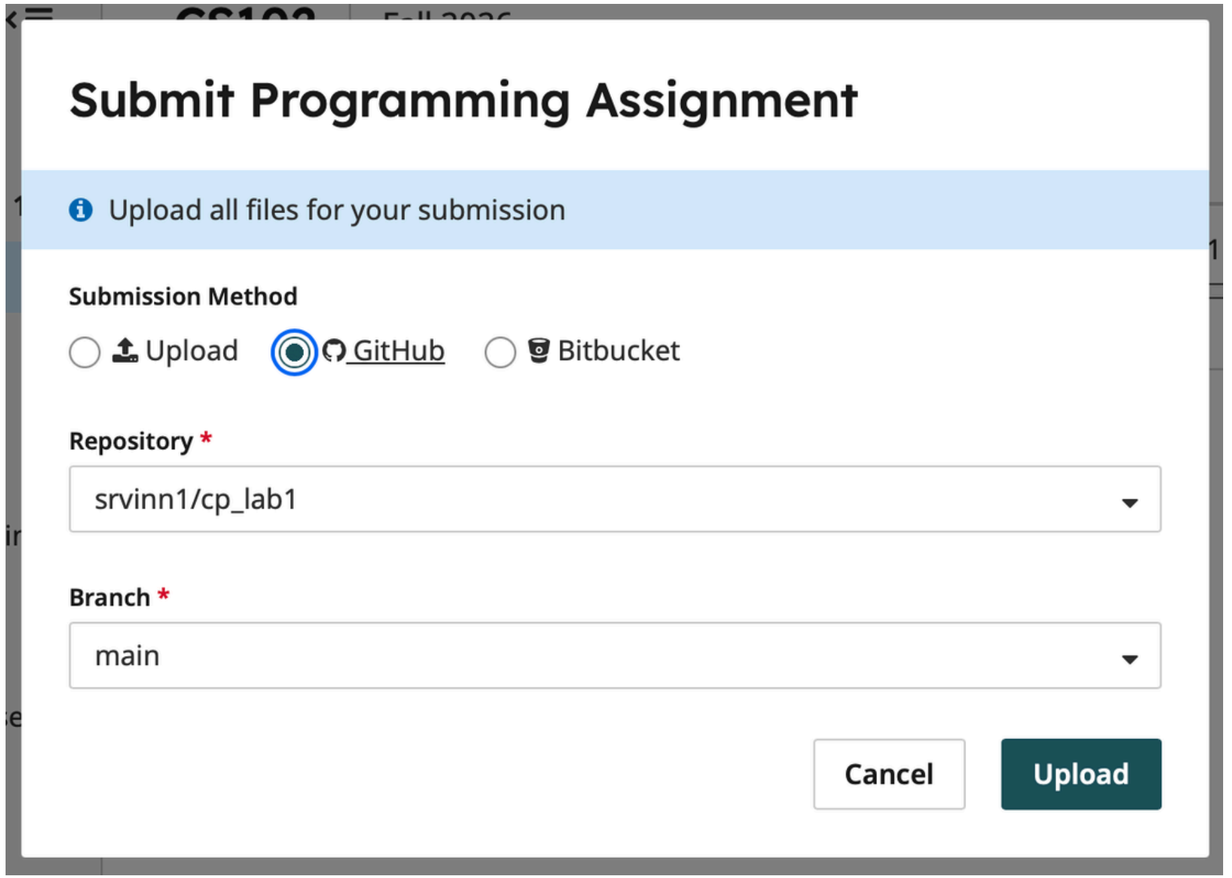
You can see all our commits!

Note that Git permanently records each commit in the repository until its history is deliberately changed or deleted.

See your commits in Github



Submit your assignment to Gradescope



That’s it!